



D3 Cell division

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Mitosis is a single division that produces two genetically identical diploid daughter cells for growth, repair and asexual reproduction; after DNA replication in interphase, the cell passes through prophase, metaphase, anaphase and telophase, with the spindle apparatus ensuring each daughter cell receives one full, identical set of chromosomes. Meiosis instead involves two successive divisions that, together with crossing over and independent assortment, produce four genetically variable haploid gametes from one diploid cell. The cell cycle is regulated by checkpoints that halt division if DNA damage is detected; failure of this control, often through mutations in genes regulating the cycle, can lead to uncontrolled cell division and tumour formation.

Worked example: follow the reasoning

Tracking chromosome and chromatid numbers through mitosis in a human somatic cell (diploid number $2n = 46$). Step 1: during interphase, after S phase DNA replication, the cell has 46 chromosomes, each now consisting of two identical sister chromatids joined at a centromere, giving 92 chromatids in total. Step 2: at metaphase, all 46 (duplicated) chromosomes align at the cell's equator, still 92 chromatids total. Step 3: at anaphase, the sister chromatids of each chromosome separate and move to opposite poles; at this instant there are 92 individual chromosomes in the cell (46 moving to each pole). Step 4: after telophase and cytokinesis, each daughter cell contains 46 chromosomes (46 chromatids, since chromosomes are now unduplicated). Conclusion: mitosis maintains the diploid chromosome number of 46 in each of the two genetically identical daughter cells.

Quick check · multiple choice

What is the main biological function of mitotic cell division?

- A To produce genetically variable gametes for sexual reproduction
- B To produce genetically identical diploid cells for growth and tissue repair
- C To halve the chromosome number from diploid to haploid
- D To generate new species through recombination

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Cell cycle checkpoints monitor conditions such as DNA integrity before allowing the cell to proceed to the next phase of the cycle. [7]
 - (a) State at what point in the cell cycle a checkpoint could prevent a cell with damaged DNA from continuing to divide. [1]
 - (b) Explain how the loss of function of genes controlling cell cycle checkpoints can lead to the formation of a tumour. [4]
 - (c) Distinguish between a benign and a malignant tumour. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) For example, at the G1/S checkpoint (before DNA replication begins) or the G2/M checkpoint (before mitosis begins), damaged DNA can be detected and the cycle halted until it is repaired or the cell is directed to undergo apoptosis.

(b) Genes that normally regulate checkpoints (e.g. tumour suppressor genes, which usually halt the cycle in response to DNA damage) can be mutated so they no longer function correctly; this allows cells with damaged or mutated DNA to continue dividing when they normally would not, bypassing the normal control on the cell cycle; repeated uncontrolled division produces a growing mass of abnormal cells, forming a tumour, and further mutations accumulating in these dividing cells can worsen the loss of control.

(c) A benign tumour grows in a confined area, often surrounded by a membrane/capsule, and does not spread (metastasise) to other parts of the body; a malignant tumour invades surrounding tissue and can spread via the blood or lymphatic system to form secondary tumours (metastases) elsewhere in the body.

Quick check answer

B. Mitosis copies and evenly distributes chromosomes, producing two daughter cells genetically identical to the parent cell, supporting growth, tissue repair and asexual reproduction.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

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